

Public Economics

Lecture 8 · Public Goods

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Today's Questions

- What makes a good “public” in the economist’s sense — and why does the market systematically undersupply it?
- If everyone has an incentive to free-ride, why does *any* private giving happen at all?
- Does government provision simply replace private giving one-for-one, or can the two coexist?

Based on

Saez, E., *Econ 131* lecture notes (UC Berkeley); Gruber, J. *Public Finance and Public Policy*, Ch. 7.

7.1 Optimal Provision of Public Goods

7.2 Private Provision of Public Goods

7.3 Crowding Out and Government Provision

7.1 Optimal Provision of Public Goods

Public Goods: Definitions

Pure public good

Perfectly **non-rival in consumption** and **non-excludable**.

- **Non-rival:** one person's consumption doesn't reduce anyone else's opportunity to consume it.
- **Non-excludable:** nobody can be stopped from consuming it.

Impure public good

Satisfies the two conditions only *partially* — most real-world “public goods” are actually this.

The economist's definition is narrow: it says nothing about whether a good *should* be universally provided. Education and health care are not public goods in this technical sense — they're rival and (at least partly) excludable — even though most societies choose to provide them broadly.

Recap: Optimal Provision of Private Goods

Two goods: ice cream (ic) and cookies (c , the numéraire, $P_c = 1$). Two individuals, B and J , face the same market price P_{ic} but may consume different quantities.

- $MRS_{ic,c} = MU_{ic}/MU_c$ = cookies a consumer will give up for one more ice cream.
- **Individual optimum:** $MRS_{ic,c}^B = MRS_{ic,c}^J = P_{ic}$.
- **Supply-side equilibrium:** $MC_{ic} = P_{ic}$.

$$\Rightarrow MRS_{ic,c}^B = MRS_{ic,c}^J = MC_{ic}$$

Each person separately equates their own MRS to the (common) marginal cost — because one more ice cream for B doesn't affect J at all.

Optimal Provision of Public Goods: The Samuelson Rule

Now replace ice cream with a public good — missiles (m) — consumed by *both* B and J **simultaneously**.

- $MRS_{m,c}^B$ = cookies B will give up for one more missile; $MRS_{m,c}^J$ likewise for J .
- Since both enjoy the same missile, society is willing to give up $MRS_{m,c}^B + MRS_{m,c}^J$ cookies for it.

Samuelson Rule (Samuelson, 1954)

$$MRS_{m,c}^B + MRS_{m,c}^J = MC_m$$

Efficiency requires marginal cost to equal the **sum** of everyone's MRS — not each person's MRS separately, as with a private good.

7.2 Private Provision of Public Goods

Private-Sector Underprovision

Left to itself, each individual i contributes only up to the point where *their own* $MRS_{m,c}^i = MC_m$ — ignoring the benefit to everyone else.

$$\sum_i MRS_{m,c}^i > MC_m \Rightarrow \text{too little of the public good is provided.}$$

Free-rider problem

When an investment has a personal cost but a common benefit, selfish individuals will underinvest.

Conceptually: private provision of a public good creates a **positive externality** on everyone else — and, exactly as with any positive externality, the market undersupplies it.

Government Provision of Public Goods

Providing public goods is the oldest function of government. The largest public goods are supplied almost entirely by the state:

1. National defense.
2. Police and justice.
3. Transportation, energy, and sanitation infrastructure.
4. Increasingly: environmental protection.

Basic science is typically government-funded, while applied R&D is mostly private; a technology becomes a genuine public good once it falls into the public domain.

A Formal Model of Free-Riding

Two identical individuals share a private good X (cookies) and jointly fund a public good F (fireworks), $F = F_1 + F_2$.

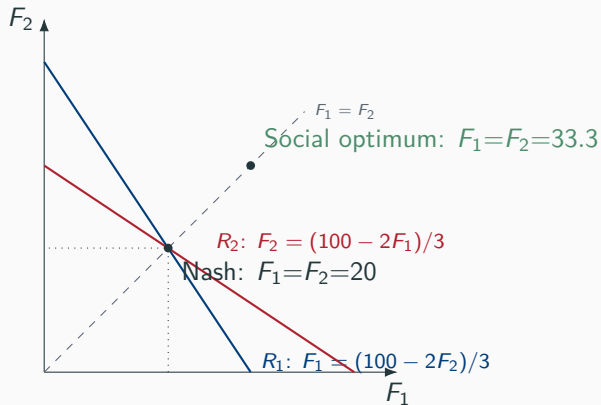
$$U_i = 2 \log(X_i) + \log(F_1 + F_2), \quad X_i + F_i = 100$$

Individual 1 maximizes $2 \log(100 - F_1) + \log(F_1 + F_2)$ taking F_2 as given:

$$-\frac{2}{100 - F_1} + \frac{1}{F_1 + F_2} = 0 \quad \Rightarrow \quad F_1 = \frac{100 - 2F_2}{3}$$

This is individual 1's **reaction curve**: F_1 falls as F_2 rises — the free-rider effect in action. By symmetry, $F_2 = (100 - 2F_1)/3$.

Diagram: Reaction Curves and the Nash Equilibrium



The two reaction curves cross at the Nash equilibrium, well *below* the socially optimal contribution on the $F_1 = F_2$ line.

Nash Equilibrium vs. the Social Optimum

At the Nash equilibrium, both reaction curves hold simultaneously:

$$F_1 + F_2 = \frac{200 - 2(F_1 + F_2)}{3} \Rightarrow F = F_1 + F_2 = \frac{200}{5} = 40 \Rightarrow F_1 = F_2 = 20$$

The social optimum requires $\sum_i MRS^i = MC = 1$, with $MRS_{F,X}^i = \frac{X_i}{2F}$:

$$\sum_i MRS^i = \frac{200 - F}{2F} = 1 \Rightarrow F = \frac{200}{3} = 66.6 > 40$$

Public goods are under-provided by the market

Free-riding cuts total fireworks nearly in half relative to the efficient level.

In-Class Game: A Public Goods Experiment

Setup

You're in a group of 5. You have 10 tokens. 1 token kept privately = \$1 for you. 1 token given to the group = \$0.50 to *each* of the 5 members (including you) — \$2.50 total.

How would you split your 10 tokens?

- A. All 10 to the private good.
- B. All 10 to the public good.
- C. Split them half and half.

Nash equilibrium (selfish play): keep everything privately. Socially optimal: contribute everything — \$1 kept is worth less to the group than \$2.50 shared.

Experimental Evidence on Free-Riding

Marwell & Ames (1981): 10 repetitions of the game above (group of 5, 10 tokens each; private return \$1, public return \$0.50 to each of 5).

- **Nash prediction:** contribute nothing.
- **Social optimum:** contribute everything.
- **What actually happens:** subjects contribute roughly **50%** to the public good.

Isaac, McCue & Plott (1985): contributions **decline** the more the game is repeated — people start out willing to cooperate, then get upset and retaliate once they see others taking advantage.

Source: Marwell & Ames (1981); Isaac, McCue & Plott (1985).

Why Do People Cooperate?

In the standard model, selfish individuals play Nash and free-ride. Yet humans plainly cooperate constantly — in families, workplaces, communities, nations.

- Cooperation appears to be **innate**, sustained by a sense of fairness and a willingness to **punish non-cooperators** even at a cost to the punisher (“altruistic punishment”).
- Likely an evolutionary adaptation.
- Fehr & Schmidt (1999) formalize “fairness preferences” in a large lab literature.

These social motives are still only partially integrated into mainstream economic models — a real limitation, especially for public economics.

7.3 Crowding Out and Government Provision

Crowding Out: A Formal Model

Suppose government now **forces** each individual to contribute 5, so $F = F_1 + F_2 + 10$, where F_i is now *voluntary* top-up:

$$U_i = 2 \log(X_i) + \log(F_1 + F_2 + 10), \quad X_i + F_i = 95$$

Solving gives $F_1 = F_2 = 15$ — **one-for-one crowd-out** of the forced contribution.

- **Why?** Let $F'_i = F_i + 5$. Then $U_i = 2 \log(X_i) + \log(F'_1 + F'_2)$ with $X_i + F'_i = 100$ — *identical* to the original (no-government) problem, so F'_i must be the same: $F'_i = 20 \Rightarrow F_i = 15$.
- Crowd-out stops being one-for-one only once voluntary contributions hit **zero** — you can't “un-contribute” a negative amount.

Crowd-out

The reduction in private contributions caused by an increase in government provision of the same public good.

Two strands of evidence:

1. **Field evidence** (observational studies).
2. **Lab and field experiments.**

Lab experiments (Andreoni, 1993) find crowd-out is real but **imperfect** — well below the one-for-one prediction of the simple model. Lab settings likely miss important real-world motives for giving: warm glow, prestige, and active solicitation by fundraisers.

Charitable giving is one form of private public-good provision.

- In the US, giving totals about **1.5% of national income** — large in absolute terms, but far smaller than the gap in government spending between the US ($\approx 30\%$ of national income) and the EU ($\approx 45\%$).
- Funds flow mainly to: religious organizations, education, human services, health, the arts, and other causes (environment, etc.).
- Governments actively **encourage** it: charitable gifts are typically deductible from taxable income.

Why Do People Give?

1. **Warm glow** — e.g., your name on a building.
2. **Reciprocity** — e.g., alumni giving back to their university.
3. **Social pressure** — e.g., giving at church, in front of others.
4. **Altruism** — e.g., poverty relief for its own sake.

None of these motives are captured by the basic free-rider model above — which is exactly why charities run large fundraising operations built around social and psychological cues, not just the pure economics of public-good provision.

Evidence: Andreoni & Payne (2003) on Crowd-Out

Government spending can crowd out private donations through **two** channels: reduced willingness to give, *and* reduced fundraising effort.

- Panel data on arts and social-service organizations' tax returns, followed over time.
- **Finding:** a \$1,000 increase in a government grant leads to about \$250 less private fundraising.

Most of the crowd-out operates through the **fundraising** channel: less government funding means organizations fundraise harder (e.g., federal funding cuts to universities tend to boost their own fundraising drives).

Crowding in

Kotchen & Wagner (2023): when the US spends more on national parks, **volunteer hours rise too** — government spending can crowd private effort *in*, not just out.

Field Experiment: Reciprocity in Giving

Falk (2007): 10,000 Christmas solicitation letters (funding schools for street children in Bangladesh) sent to Swiss donors, randomized into three groups.

Group		Gift included	Response rate
Control		None	12%
Treatment 1	Small (1 postcard + envelope)		14%
Treatment 2	Large (4 postcards + envelopes)		21%

A larger upfront gift substantially raised giving — consistent with reciprocity, and effective even relative to its cost.

Source: Falk (2007), *Econometrica* 75(5).

Field Experiment: Social Pressure in Door-to-Door Giving

DellaVigna, List & Malmendier (2012): a randomized door-to-door fundraiser experiment.

- **Control:** no advance warning of the visit.
- **Treatment 1:** a doorknob flyer announces the exact visit time (so residents can seek out — or avoid — the fundraiser).
- **Treatment 2:** same flyer, plus a “Do not disturb” checkbox.

Results (vs. control): Treatment 1 — 9–25% less likely to open the door, unconditional giving unchanged. Treatment 2 — many opt out, and unconditional giving falls 28–42%. Social pressure is a real determinant of door-to-door giving — and the fundraiser’s visit itself *lowers* the utility of potential donors.

Social Prices as a Policy Instrument

Beyond taxes and subsidies, policy can try to shift **social norms** directly — “social prices.”

- Make a positive-externality behavior a status symbol: electric cars are increasingly admired, while large gas-guzzling SUVs are increasingly frowned upon (Butera et al., 2022).
- Social-price elasticities can be large: Gerber, Green & Larimer (2008) show that a letter using social pressure (“your neighbors will see whether you voted”) substantially raises voter turnout.

Should social pricing be layered *on top of* standard corrective taxes or tradable permits?

- **Shaming SUV drivers** is generally **inefficient** relative to a gas tax: it destroys welfare without raising any revenue. It can still be worthwhile when a fine or tax is hard to enforce at all — e.g., social disapproval of littering, where fines are difficult to levy.
- **Praising efficient-car owners**, by contrast, is **efficient** relative to a gas tax: it adds to welfare, since driving the energy-efficient car becomes more enjoyable in itself.

Andreoni, J. (1993), *AER*.
Andreoni, J. & Payne, A. (2003), *AER*.
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Thank you!

Questions and discussion